Probability, R and Computational Finance

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Introduction

> "What I hear, I forget; What I see, I remember; What I do, I understand"
>
> — Confucius, 551-479 BC

Our lectures are going to teach you basic concepts of probability and their applications in computational finance. Our approach is based on intuition and on understanding concepts by building them with your own hands.

But how can we build abstract concepts such as probability, random phenomena and chance by our own hands? How can we "touch" such things like an idea? While probability is a mathematical theory, it gains practical value and an intuitive meaning in connection with real or conceptual experiments such as, tossing a coin, rolling two dice, playing roulette, observing the change in the index value of stocks, bonds or in exchange rates, selecting a random sample of financial transactions in a payment system and observing the number of transactions with a value below a certain amount.

Many of these experiments we can nowadays do not only physically – by actually rolling a pair of dice, for example – but we can use the computer to build a virtual dice and roll it on the computer. We can build and simulate a huge variety of random phenomena. We can - for instance - implement models of random fluctuations of asset prices. We can model financial risks and contemplate possible future scenarios through simulation.

We can approach our understanding of abstract concepts by building them with our own hands on the computer. This is the approach to teaching you probability in this course.

There are many ways to build virtual objects and to run simulations to manipulate them. For this we will need a programming language. The language we choose for this course is R and the integrated development environment RStudio. This is one of the main languages used in data analysis, statistics and data science and is widely used in industry and academia. It will be our tool to do probability in this course.

Downloading and installing R

So lets start by downloading and installing R first. R is an open source project maintained by an international team of developers. The software is made available through a website called the comprehensive R archive network (http://cran.r-project.org).

At the top of this website, in a box named "Download and install R" you will find three links for downloading R. Choose the link that describes your operating system, Windows, Mac or Linux. These links will lead you to the necessary information you need to install a current version of R. The easiest install option is to install R from precompiled binaries. There is also the option to built R from source on all operationg systems if you have the tools and the expertise to do so. R also comes both in a 32-bit and a 64-bit version. It does not make a substantial differecne which version you use. 64-bit versions can handle larger files and datasets with fewer memory management problems.

Downloading and installing RStudio

RStudio is an application that helps you write and develop R code. It makes using R much easier for you than using it in isolation. The interface of RStudio looks the same across all operating systems.

You can download RStudio for free from https://www.rstudio.com/products/rstudio/. Select the box RStudio Desktop and follow the download instructions. RStudio Desktop is free. Note that you need to have a version of R installed to use RStudio.

If you have successfully installed R and RStudio, we are ready to start.

Empirical background and some terminology

If you chose to study Finance it should be natural for you to study probability. After all, Finance studies the allocation and pricing of money across time. You can safe money to use it later for purchasing goods and services you can borrow money to invest today and repay in the future from the revenues of your project. But the future is uncertain and we do not know today how it will look like. In Finance we unavoidably have to deal with risk and uncertainty. The most fundamental concept we have available to deal with risk and uncertainty and to think about and analyze it is probability. Therefore every serious student of Finance has to study probability early on.

Terminology

Probability is a mathematical theory. This theory gets practical value and intuitive meaning in connection with real or conceptual experiments such as rolling a dice, tossing a coin or selecting a random sample of stocks and observing their most extreme price drop over a given period of time or observing the frequency of car accidents etc. All of these descriptions are, however, rather vague. We have to be more precise what we mean by an experiment or observation.

When we use the term (random) *experiment* we mean a process leading to an uncertain outcome. To make this notion precise and useful we need to agree what we mean by uncertain outcomes.

Take the simple example of tossing a coin. In a practical situations the outcome of a coin toss can be that the coin lands on head or tail. But it could in principle also fall on an edge, stand and roll away. Still when we think about the experiment of tossing a coin in the theory of probability we agree *on the outset* that heads and tail are the only possible outcomes of this experiment. By such an agreement we have pinned down the possible outcomes.

The collection of all possible outcomes of an experiment is calles the *sample space*. While the sample space of the experiment of tossing a coin is an idealization

it is exactly this idealization which simplifies the theory without affecting its applicability.

Idealizations of this kind are standard in probability and are widely used also in Finance. It is - for example - impossible to know how high the price of a stock might become. Is there a maximal price beyond which the price can not possibly hike? Many of us would hesitate to claim that the price might rise without bound. Yet many models in applied Finance are based on such an assumption. The models allow arbitrary price hikes but with arbitrary small probability as the price gets higher and higher. Practically it does not make sense to believe that a security price can become arbitrarily high. The use of arbitrarily small probabilities in a financial model might seem absurd but it does no practical harm and makes the model simple and convenient to use. Moreover, if we seriously introduced an upper bound on a security price at x it would be also awkward to assume that it is impossible that it could be just a cent higher, an assumption equally unappealing than assuming it can get in principle arbitrarily high.

When we talk of a *basic outcome* we mean the possible outcome of a random experiment. An *event* is a subset of basic outcomes. Any event which consists of a single outcome in the sample space is called a *simple event*.

Probability is a measure of how likely an event of an experiment is. Thus when we talk about probability in a precise and meaningful sense we can only do so in relation to a given sample space or to a certain conceptual experiment.

Thus before we can begin to seriously talk about probability we need to develop a good understanding of experiments, sample spaces, basic outcomes and events. In the spirit of this course we will develop this understanding by building these concepts using R. Thus let us develop examples of random experiments, sample spaces, basic outcomes and events in parallel with making our first steps in R. R will be our tool which will allow us to "touch" probability concepts.

As we proceed through our course we will develop a broader and deeper understanding of this language together with the probability concepts.

Rolling a die: Constructing a random experiment, a sample space and events using R

Lets start with a classic and old example of a probability model: Rolling a six sided die. We want to roll this die not physically but on our computer using R and RStudio as our tool. How can we do this?

The R User Interface

Before we can ask our tool to do anything for us, we need to know how to talk to it. In our case RStudio allows us to talk to our computer. It works like any

other application. When you launch RStudio on your computer, you will see a screen looking like this:

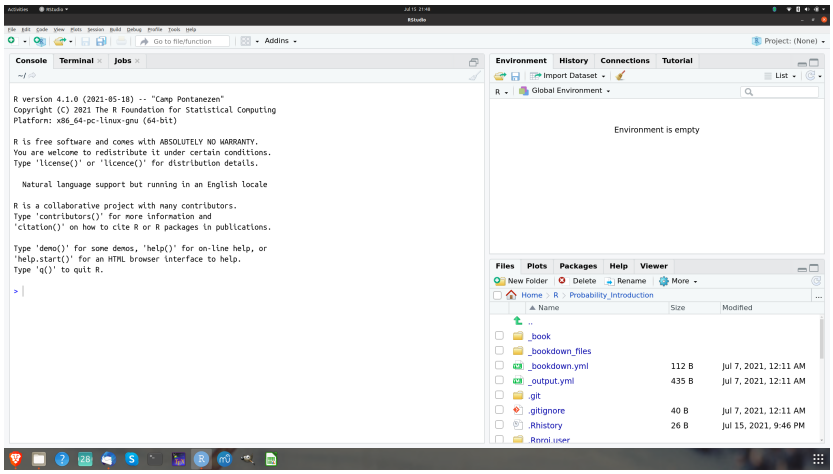

Figure 1: The RStudio startup screen

In this picture you see a screenshot of my RStudio screen. Interacting with the app is easy. You type commands via your keyboard at the prompt, which is the `>` symbol. You can see it in the left pane in the screenshot. You send the command to the computer by pressing enter. After you have pressed enter, RStudio sends the command to R and displays the result of your command with a new prompt to enter new commands, like this:

```
> 1 + 1
[1] 2
>
```

The `[1]` means that the line begins with the first value of your result. For example, if you enter the command `20:60` which means in the R language, "list all the integers from 20 to 60," and you then press return, you will get an output like this:

```
[1] 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
[34] 53 54 55 56 57 58 59 60
```

meaning that 20 is the first value displayed in your result. Then there is a line break because not all values can be displayed on the same line and R tells you that `53` is the 34-th value of the result.

The colon operator `:` is a very useful function in R which we will often need. It allows us to create sequences of every integer between two given integers.

R needs a complete command to be able to execute it, when the return key is pressed. If a command is incomplete, like for instance `5*`, where we enter 5, the

multiplication operator but no second factor to multiply 5 with R will show:

```
> 5*
+
```

The + sign means that R needs more input to complete the expression. So if we type the missing factor after the + prompt, say 4 and then press the return key again, we will get the result

```
> 5*
+ 4
[1] 20
```

If you type a command that R does not understand, you will be returned an error message. Don't worry if you see an error message. It just is a way the computer tells you that he does not understand what you want him to do.

For instance, if you type 5%3 you will get an error message like this

```
> 5%3
Error: unexpected input in "5%3"
>
```

Sometimes it is obvious why a mistake occurred. In this case, that R just does not know what to do with the symbol %. It has no meaning in this context. Sometimes it is not so obvious what the error message actually means and what you might do about it. A useful strategy in this case is to type the error message into a search engine and see what you can find. The chance is very high that others encountered the same problem before you and got helpful advice how to fix it from other users on the internet. One site, we find particularly helpful for all kinds of questions related to R and R programming is https://stackoverflow.com/. Try it at the next opportunity.

Now with this basic knowledge, we can already make the first step to create a die on the computer using R. If you think of a physical die, the essential thing that matters are the points on its six sides. If you throw the dice it will usually land on one of these sides and the side which is on the upper side of the die shows the number of points. The colon operator : gives us a way to create a group of numbers from 1 to 6. R gives us the result as a one dimensional set of numbers.

```
> 1:6
[1] 1 2 3 4 5 6
>
```

Lets use these first steps in R to recap the probability concepts we have learned using this example of the six sided die: A *basic outcome* of rolling a six-sided die is for example 6 if the upper side after rolling the die happens to be the side with 6 points. The *sample space* of the experiment of rolling a six-sided die is the set $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. In probability theory we often use the symbol $\mathcal{S}$ or S for sample space. In many probability texts the sample space is also often denoted

by the symbol Ω the greek letter for (big) Omega. A *random experiment* in this example is the rolling of the die. The outcome is uncertain but once the die is rolled the outcome can be determined precisely. Finally we can also illustrate in the example the concept of an *event*: For example, the event, that the die will show less than 4 points is $A = \{1, 2, 3\}$. We use here the symbol A to denote this event and use set notation to describe the set of basic outcomes that make up the event "number of points smaller than 4." The event that the outcome is that the point shows 10 points is the empty set $B = \emptyset$. The symbol $\emptyset$ comes from set theory and means the set containing no elements. This event can contain no elements because we can not get a score of 10 by rolling a six sided die.

Objects

You can save data in R by storing them in *objects*. An object is a name, you can choose yourself to store data. For example, if you choose to store the value `6` in an object called `point_six`, you would type:

```
> point_six <- 6
>
```

at the prompt. R will the store the value 6 in the object called `point_six`, which you can use to refer to the value. If you type the name of your object at the prompt, R will distplay the value you have assigned. A usful key combination for typing the assignment operator `<-` is to use the key combination `ALT _`. At the R prompt R will automatically print an assignment operator.

```
> point_six
[1] 6
>
```

Now you can use the name of the object to refer to its value. For instance, you could divide `point_six`by 2and get a meaningful result

```
> point_six/2
[1] 3
>
```

Now to make our `die` more tangible and useful, let us store it in an R object by typing the following command at the prompt. This command creates an object with name `die` and assign the vector `1,2,3,4,5, 6` to it.

```
> die <- 1:6
>
```

You can now see in the right upper Environment pane that R shows you that there is an object with the name `die` that it consists of integers `1,2,3,4,5`. As you create more objects they will be stored in the Environment pane and are ready for your reference, unless you delete them. You can remove or delete an object by typing

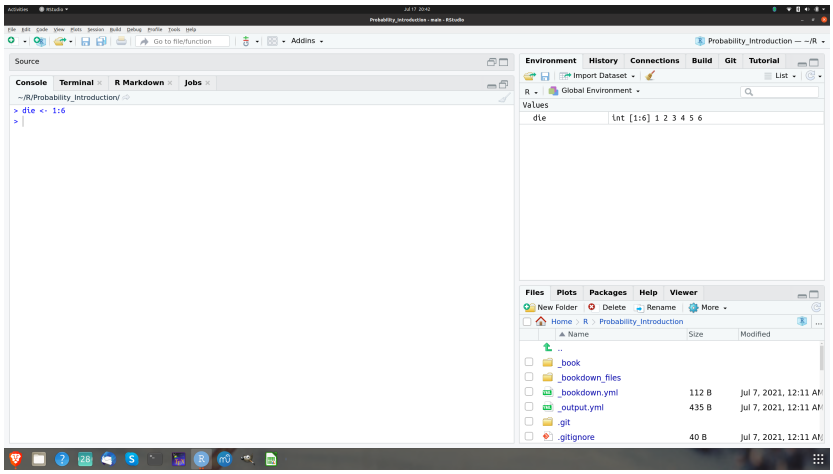

Figure 2: The RStudio Environment pane keeps track of the objects you have created

```
> rm(object)
>
```

or by assigning the value `die <- NULL` which would also remove the object from your environment or workspace.

You can name your objects almost anything with a few exceptions. An object name must not start with a number. There are some special symbols which can also not be used in object names, like `^, !, $, @, +, -, /, *`. Note that R is case sensitive and distinguishes small and big letters. If you assign a new value for an object you have already created, R will overwrite the object without warning.

You can see which objects are currently created and available for you in the Environment pane of your session of by typing `ls()`.

Before we learn how we can actually roll our die and perform a random experiment with it, let us briefly use the opportunity to explain a few things about how R does computations. We have already explained that we can use the object name to refer to the value. So for instance if we type

```
> die*die
[1]  1  4  9 16 25 36
>
```

This might irritate some of you. we have called the object a vector. In linear algebra multiplication of vectors is only allowed if there is an inner product. What happens here, if we use `*` the multiplication operator is that R does an element-wise multiplication of the six numbers of our die. Of course R alows to

take an inner product as well, but this needs a different operator. To compute an inner product, we would type

```
> die %*% die
     [,1]
[1,]   91
```

Now R displays the result as a vectors with one row and one column, which is denoted in the output by `[ , 1]` for the column and `[1, ]` for the row. We will learn later more about the use and the meaning of this notation in R.

The elementwise execution R usually uses also means that when you, for example type

```
> die - 1
[1] 0 1 2 3 4 5
>
```

R would subtract `1` from every component in the vector `die`.

Another specific behavior of R, you need to know about is called "recycling." If you give R two vectors of different length in an operation, R will repeat the shorter vector as long as it is of equal length with the longer one. For example, if you have:

```
> die + 1:2
[1] 2 4 4 6 6 8
>
```

you see that R adds 1 to 1 and 2 to 2 and then starts over again by adding 1 to 3 and 2 to 4 and then starts over once again by adding 1 to 5 and 2 to 6. If the longer vectors is not a multiple of the shorter one, R recycles but the ciuts off. It will give you a warning though.

```
> die + 1:4
[1] 2 4 6 8 6 8
Warning message:
In die + 1:4 :
  longer object length is not a multiple of shorter object length
>
```

While this might seem awakward to some of you, we will see that for data manipulation element-wise execution is often extremely useful. It allows to manipulate groups of values in a systematic way.

Functions

R contains many functions whcih we can use to manipulate data and compute things. The sytax for usuing a function is very simple: You type the function name and put the value of the function argument in parentheses. Here we use for

illustrations the function of the squareroot `sqrt()` and the function for rounding a number `round()`.

```
> sqrt(4)
[1] 2
> round(3.1415)
[1] 3
>
```

The data you write in the parentheses are called the fuction arguments. Arguments can be all sorts of things: raw data, R objects, results from other functions. If functions are nested, R evaluates the innermost function first and then goes on to the outer functions. To see examples of all these instances you can take

```
> mean(1:6)
[1] 3.5
> mean(die)
[1] 3.5
> round(mean(die))
[1] 4
>
```

for example.

For simulating random experiments, R has the function `sample()`. With this function we can roll our die on the computer and conduct actual random experiments. The function takes as arguments a vector names `x` and a numer named `size`. `sample` will return `size` elements randomly chosen from the vetcor `x`. Lets say:

```
> sample(x = 1:4, size = 2)
[1] 4 1
>
```

In this case `sample` has chosen `4,1` from the vector `x = (1,2,3,4)`. If we want to roll the die in our computer we can thus pass the die as an argument to `sample` and use the number `1` for the `size` argument. Lets do a few rolls with our die

```
> sample(x = die, size = 1)
[1] 3
> sample(x = die, size = 1)
[1] 2
> sample(x = die, size = 1)
[1] 3
> sample(x = die, size = 1)
[1] 5
> sample(x = die, size = 1)
[1] 1
> sample(x = die, size = 1)
```

```
[1] 5
>
```

R functions can have many arguments, but they need to be separated by a comma.

Every argument in every function has a name. We specify which data are assigned to the arguments by setting the name equal to the data. Names help us to avaoid passing the wrong data and thereby mixing up things or committing errors. But using names is not necessary. If we just wrote

```
> sample(die, 1)
[1] 5
>
```

R would also know what to do. It is not always clear which names to use for a function. If you are not sure, you can use the function `args` to look it up. Here we take the function `round` as one example.

```
> args(round)
function (x, digits = 0)
NULL
>
```

Note that the `digits` argument in the `round` function is already set to `0`. Frequently R functions come with optional arguments. These arguments are optional because the come with a default value, which is `0` in case of the round function.

We recommend that you write out argument names as a rule. It gives clearer code and avoids errors. If you don't write argument names, R matches your values to the arguments of the function by order.

Writing your own functions

Now we are ready to write our own function to roll the die in our computer. Each function in R has the same elements: A name, a function body of code and a set of arguments. To write your own function, you have to write up all of these parts and save them in an R object. The syntax is:

```
my_function <- function() {}
```

The name here is `my_function`, next comes the expression `function()` which needs to be assigned. The names of the function arguments have to be written between the parentheses. Then we have to write the actual code within the braces `{}`.

To do this for the die, lets write a function names `roll_die`.

```
> roll_die <- function(){die <- 1:6
```

```
sample(die, size = 1)}
```

Now we can roll our die for a few times to show how the function works

```
> roll_die()
[1] 4
> roll_die()
[1] 1
> roll_die()
[1] 5
> roll_die()
[1] 4
> roll_die()
[1] 6
>
```

A final remark in the `sample`function is in place here. If we look at the argunments of sample using the `args` function we see

```
> args(sample)
function (x, size, replace = FALSE, prob = NULL)
NULL
>
```

Lets do not discuss all the details of this output but concentrate for a moment on the `replace` argument. What does this mean? As we saw previously we can use the sample function to model the rolling of our die. If we set the `size` argument to `1` we get the roll of one die. If we set the `size` argument to `n`, we would simulate the rolling of `n` dies. But now the `replace` argument becomes crucial. As we can see in the output of the `args` function `replace` has a default value `FALSE`. This is a logical argument. It tells R, for example, that if we set `size = 2`, meaning that two dice are rolled, if the first dice shows, say a value of 3, the second die cannot show 3 as well. This is clearly not what we have in mind when we model the rolling of 2 dice. It should be possible that both dies show the same value. To enable this behavior of the sample function, we have to change the default value of the `replace` argument to `TRUE`. Then R chooses a random draw from all of the six possible values for all dice rolled.

Congratulations. You have written your first R function for conducting a simple random experiment. Think of the parentheses as a trigger that tells R to run the function. If you omit the trigger R just prints the body of the function. When you run a function, all the code in the function body is executed and R returns the result of the last line of code. If the last line of code does not return a value neither will R.

Arguments

Imagine we remove the first line of code in our function body and changed the name die in the sample function to ball.

```
roll_die2 <- function(){sample(ball, size = 1)}
```

If we call the function now, we will get an error

```
> roll_die2()
Error in sample(ball, size = 1) : object 'ball' not found
```

We could supply ball when we call "'roll_die2"" if we make ball an argument of the function. Lets do this:

```
> roll_die2 <- function(ball){sample(ball, size = 1)}
>
```

Now the function will work as long as we supply `ball` when we call the function.

```
> roll_die2(ball = 1:6)
[1] 2
>
```

Note that we still get an error, if we forget to supply ball argument

```
> roll_die2()
Error in sample(ball, size = 1) :
  argument "ball" is missing, with no default
```

This could be avoided if we give the function a default argument

```
> roll_die2 <- function(ball= 1:6){sample(ball, size = 1)}
> roll_die2()
[1] 4
>
```

Scripts

So far we have worked by interacting with the console. But what if you want to edit your functions? It would be much easier, if you could use a draft of your code and work form there. This can be done by using a *script*.

You create a script by going to `File > New File > R script` in the menu bar of RStudio. Using scripts is the stander way to write code in R. It not only helps you to keep track of your code, save it and edit it later. It also makes your work reproducible. You can edit and proofread your code and share it with others. To save your script go to `File > Save As` in the menu bar.

RStudio has many useful features to help you work with scripts. You can for instance automatically execute a line in a code by using the run button. You can also execute sections of code or the entire script. The entire script is executed

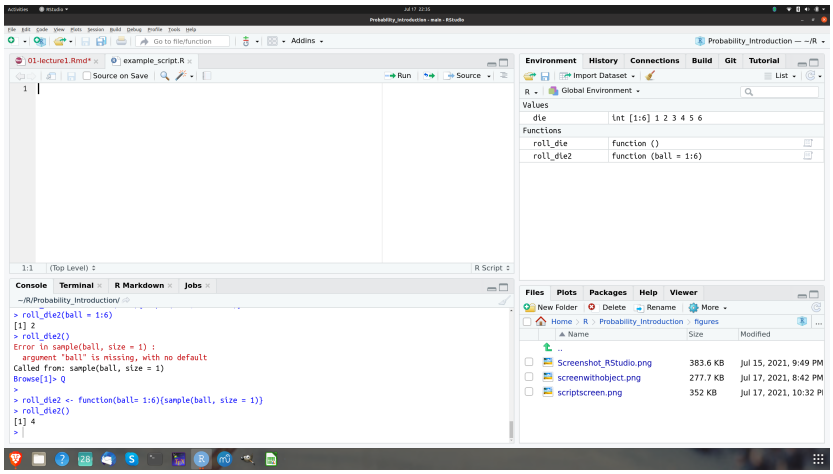

Figure 3: The RStudio Script

by running the Source button. For all these commands there are key short cuts which you will learn as you work more with RStudio and R.

Using packages and finding Help

We have now a function which we can use to simulate the rolling of a die, `roll_die()`. If the die is fair it should be the case that if we roll the die often, all numbers should occur about equally often. The die should not be weighted in favor of a particular value.

One way to learn whether our die is fair are *repetition* and *visualization*. These are tools we will need all the time, when working with data and when doing probability. While R has many useful functions, one of the great powers is that R is contsnatly extended by a huge community of users by providing packages. Packages are add on functions, which will not be available when you install R. They need to be installed and loaded before you can use them. Since packages are such a powerful tool in R we need to introduce what they are and how to use them in the beginning.

Packages

There are many visualization tools in R that come with the basic installation. Since the point we want to make here is about packages, we will use a visualization function which is part of the package *ggplot2*, a very popular package for making all kinds of graphs. There are many additional functions provided with this package. For the point we want to make here we will use one of them, called `qplot`.

Since `qplot` is a function in the package *ggplot2*, we first need to install this package. To install a package you need to be connected to the internet. If you have internet connection go to the command line and run at the command line

```
> install.packages("ggplot2")
```

R will now download the package via the internet and install it. It will display various messages in the console during this process. Don't worry if you don not know what they exactly mean and don't panic that they are displayed in red. All packages can be installed like this. You have just to enter the correct name in the function `install.packages()`. The lower left pane in the RStudio software alternatively provides a tab called **Packages**. Using this tab, you can also install R packages interactively by clicking the install button on the upper left corner of the Packages tab.

After installation the package is on our hard-drive but it can not yet be used. To use the package it has to be laoded. This is done with the command `library`. To load the *ggplot2* package we type

```
> library("ggplot2")
```

and hit the return key. Many things could be said about the R package system and you will learn it in more detail as we go along in our course. For the moment the most important thing to remember now is that a package needs to be newly loaded whenever you want to use it in a new R session.

To check whether our dies is fair, we need to roll it many times. R provides a function, that does this for us. This function is called `replicate` and provides an easy way to repeat a command many times. The number of times we want to repeat something is given as an argument to the `replicate` function.

Now lets roll our die 100 times and save the result in an object we call `rolls`:

```
rolls <-replicate(100, roll_die())
```

We now use the "plot"' function from the *ggplot2* library to make a quick visualization.

It looks like every value occcured roughly 16 times but there is still quite some variation. For instance, the value 3 seems to occur more than 20 times whereas the value 2 occurs less than 15 times. Maybe we have to give it another trial with more replications. With the computer we can do this with a fingertip. Let us roll our die 10000 times and plot the result.

```
> rolls2 <- replicate(10000, roll_die())
> qplot(rolls2m bindwidth = 1)
```

Now the picture looks better.

It seems that there is no significant visual evidence that our die is loaded.

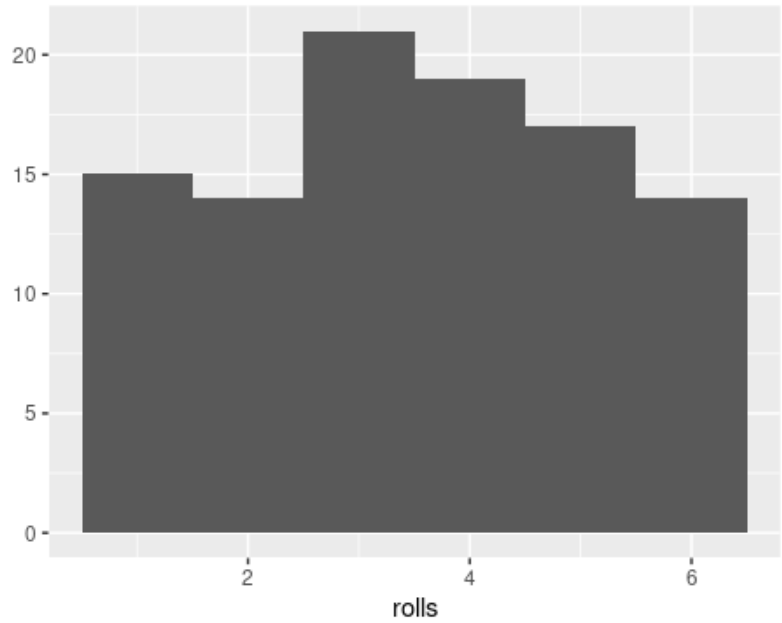

Figure 4: Frequencies of 1, 2, 3, 4, 5, 6 after rolling our virtual die 100 times

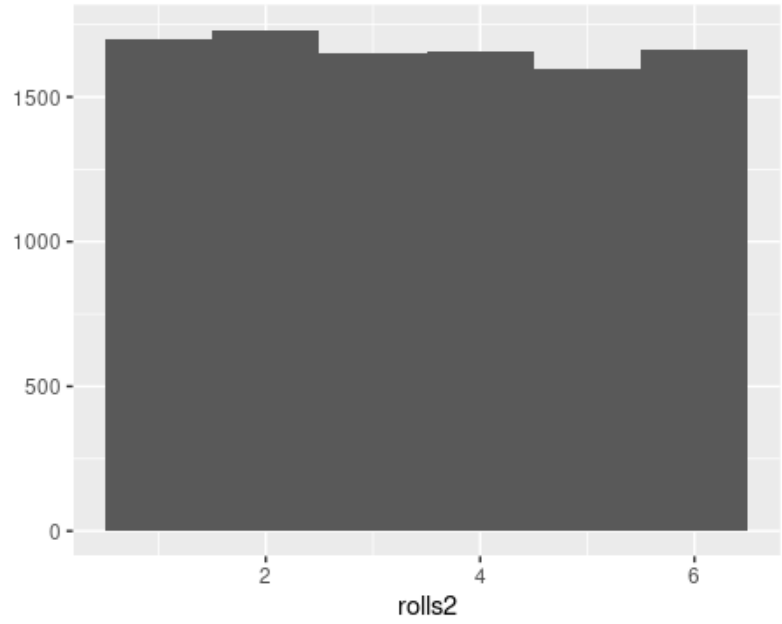

Figure 5: Frequencies of 1, 2, 3, 4, 5, 6 after rolling our virtual die 10000 times

Getting Help

We have now learned a tiny number of R functions and we have written one function ourselves. We have learned how to make use of functions provided by packages. It would be overwhelming to memorize and learn them all. In R, fortunately, every functions comes with a detailed documentation and with its own help page. You need to learn how to use this source right from the beginning.

To access the help page of a function you type the function name preceeded by a questionmark at the prompt, like this

```
> ?sample
```

Then, after you have pressed the return key, a help page is opened in the right lower pane under the help tab. This help page has a particular structure that you will find for every other R function no matter whether it is provided by the base installation or by a package. In the upper left corner you see the name of the function (`sample`) and in curly brackets next to it the term `base`, which means that this is a function in the R base installation. Then you see a headline about what the function does.

From the top of the page, you then first see the field **Description**. This is a short description what the function does. Here it says

```
Description
sample takes a sample of the specified size from the elements of x using either with or without r
```

The next field is **Usage**. It gives you the function description with the arguments. Here for example

```
Usage
sample(x, size, replace = FALSE, prob = NULL)

sample.int(n, size = n, replace = FALSE, prob = NULL,
           useHash = (!replace && is.null(prob) && size <= n/2 && n > 1e7))
```

The first line in Usage should by now be familiar. Don't worry about the second line. The function can obviously do more than we know so far.

Next comes a list of arguments the function takes and what type of information R expects you to provide, as well as what R will do with this information. Here it says for example

```
Arguments
x
either a vector of one or more elements from which to choose, or a positive integer. See ‘Details

n
```

```
a positive number, the number of items to choose from. See 'Details.'

size
a non-negative integer giving the number of items to choose.

replace
should sampling be with replacement?

prob
a vector of probability weights for obtaining the elements of the vector being sampled

...
```

We omit some information here.

Then comes a field called **Details** which gives a more in-depth description of the function. The next field is calles **Value**. It describes what the function returns when you run it. Then we have a referecne to related R functions under the field **See**. Finally there is a field called **Examples**. This field contains example code that is guaranteed to work. It shows a couple of different cases how you can use the function in practice.

Some more examples

To close this lecture let us come back to proabability and discuss some more examples of random experiments, sample spaces, basic outcomes and events. Let us try to leverage our R knowlegde to build the objects in the different examples ourselves.

Tossing a coin twice

Suppose we want to study the random experiment of tossing a coin twice. This is a random experiment because whether the coin will land on heads (H) or tail (T) is uncertain before we make the throws, but once we have made the throws, the outcomes are clear. The sample space is $\mathcal{S} = \{HH, HT, TH, TT\}$. The event that heads will occur at the first throw of the coin is $A = \{HH, HT\}$. The event that tail does not occur is $B = \{HH\}$, which is also a basic outcome of the random experiment.

While this example is so simple that we do not need a computer, let us take the opportunity to introduce a new R function, which computes all possible combinations of objects. Let us first create a virtual coin which can land either heads or tails.

```r
coin1 <- c("H", "T")
coin2 <- c("H", "T")
```

There are some new things here. Let us explain. First, while we used gray boxes to show you input and output of code as it would appear if you interact on the console we use here a very convenient feature in RStudio, which we will use from now on. We compose these lecture notes in a document format, which allows the mixing of text and code. Whenever we write a piece of R code, it is shown in a gray box, as before but without the `>` prompt. When you move the cursor on the gray box now, you will see a copy icon in the upper right corner of the box. This will allow you to copy this code snippet and use it in another document or on your console to execute it.

The second new thing is that we have used a new data type, which we will need very often: characters. Characters can be used to form strings and R has many powerful functions to work with and operate on strings. We will see many examples of this later. Strings are always between `" "`, like this. Like numbers, strings can be stored in objects and referred to later. But what we need here is actually two strings in one vector. This is achieven by using the `c()` function of R. c stands for concatenation and here it stores both basic outcomes of a coin throw in two single vectors `coin1` and `coin2`.

We can now pass these two virtual coins to the R function `expand.grid()`. This function will produce the sample space of two coin throws by computing all possible combinations of sequences of heads and tails that can occur in two throws.

```
expand.grid(coin1, coin2)
```

```
##   Var1 Var2
## 1    H    H
## 2    T    H
## 3    H    T
## 4    T    T
```

Don't worry at the moment if you do not understand what the output format is and why the columns are headed by **Var1** and **Var2** followed by the symbols `<fctr>`. We will come back to this.

Before we look at yet another sample, let us mention that this coin tossing setup, simple as it may seem, plays a very important role in the theory of random price fluctuations of asset prices and in option pricing and we will encounter it later in detail.

Boys and Girls

A family has four children. Which one of these children will turn out to be a boy or a girl is uncertain ex ante but once the baby is born we can see whether it is a boy or a girl. Conceptually we can thing of this as a random experiment. Can you figure out the sample space ? Can you figure out the event that the family will have two boys?

Birthdays

Assume there are 10 students in a class. When we assume that the year has 365 days, how many configurations of birthdays are possible? Again we can conceptually think of the birthday of the students in the class as the outcome of a random experiment. The number of basic outcomes are all the possible configurations of 25 birthdays among the 365 days of the year.

Here we encounter a situation where it would be possible but impractical to write down the sample space explicitly. It would just be a very long list of 25 dates long sequences of birthdays. Not very useful. Most of the time sample spaces will be like this. It is therefore practical to know counting techniques to figure out the number of possible outcomes. The example we are considering here is a special case of a general counting problem often encountered in probability: The random placement of `r`` balls inncells. The mathematical notation for such a counting problem is: $\binom{n}{r} = \frac{n!}{k!(n - r)!}$ The left habd side of the equation means "the number of ways to choose r things from among n things. The right hand side shows a formular how this can be computed. R has a function to compute these things directly. It is called`choose()‘ and we apply it to the birthdays example

```
choose(365, 10)
```

```
## [1] 1.021296e+19
```

A pretty large number. Here you see that R displays the number in scientific notation meaning $1.021296 * 10^{19}$, a number with 13 zeros. Pretty large.

Summary

These are the basic **probability concepts** we have covered in this lecture:

1. A **random experiment**: A process leading to an uncertain outcome.
2. **State space**: The collection of all possible outcomes of a random experiment.
3. **Basic outcome**: A possible outcome of a random experiment.
4. **Event**: An event is a subset of basic outcomes. Any event which contains a single outcome is called a simple event.

Project 1: Rolling two dice and adding the points

Here we formulate a project where you can apply the concepts we have learned in this lecture. In this project we ask you to write an R function that virtually throws two dice and adds up the number of points shown by the dice. Using this function do a simulation of this random experiment. Install and load the

`ggplot2` package to make a plot using the qplot function to visualize the behavior of our dice. Are the dice fair?

Probability: Basic Definitions and Rules

Probability in theory and applications of probability

Now that we have introduced the concepts of a random experiment, sample space, events and basic outcomes, we can finally proceed to introduce the notion of *probability*. From the preceding lecture it should be clear that we always speak of probability in relation to a given sample space or a conceptual random experiment.

In lecture 1 we said that *probability* is a measure of how likely an event of an experiment is. Probabilities are expressed as numbers. As observed in Feller (p. 19), these numbers are of the same nature as distances in geometry. In the theory we assume they are given to us. We need not assume anything about how they are measured. In actual applications, the determination of probabilities or the application of the theory to observations requires often sophisticated statistical methods. So, while the mathematical as well as the intuitive meaning of probability are clear only as we proceed with the theory we will get a better ability to see how we can apply this concept.

Basic rules of probability

When we talk of probabilities in this lecture we mostly consider probabilities with relation to a *discrete* sample space. The simple sample spaces we considered in lecture 1 contain only a finite number of points. Such sample space belong to the discrete sample spaces. There are also more complicated discrete sample spaces: Think of the random experiment of tossing a coin as often as necessary to see Heads for the first time. We can begin writing down the basic outcomes as: $E_1 = H, E_2 = TH, E_3 = TTH, E_4 = TTTH, ...$ We may consider it possible that Head never appears. This event we should perhaps call E_0. In this case, when the basic events can be arranged into a simple sequence. A sample space is

called *discrete* if it contains only finitely many points, or infinitely many points which can be arranged into a simple sequence. Not all sample spaces are discrete. Except for the technical tools required there is no essential difference between the two cases. In our discussion of probability in this lecture we consider only discrete sample spaces.

Given a (discrete) sample space $\mathcal{S}$ the probabilities assigned to events in this sample space must always fulfill three rules:

1. $P(\mathcal{S}) = 1$, where $\mathcal{S}$ is the sample space.
2. For any event $A \in \mathcal{S}$, $0 \leq P(A) \leq 1$. The probability of an event can never be negative or larger than 1.
3. The probability of the union of two events A and B is always smaller or equal to the sum of the probability of these events looked at in isolation: $P(A \cup B) \leq P(A) + P(B)$.

In the theory of probability we use the language of sets and set theory to describe relations among event. We have used one of these relations in rule three, where we expressed the relation between two sets as their union $\cup$. It is useful in studying probability to know a few of these set theoretic definitions.

Let's go through them and illustrate the concepts in the context of the examples we have already developed in lecture 1.

Union The **union of two events** A and B is the set that contains all events that are either in A or in B or in both sets. Set union is written as $A \cup B$.

Let us use the example of the experiment of rolling a dice. The sample space $\mathcal{S}$ is the set of all possible outcomes of rolling the dice $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Assume one event is that the outcome is 1, 2 or 3. In set notation, we would write $A = \{1, 2, 3\}$. Let us also assume that the second event is that the outcome is some even number, i.e. 2, 4, 6. Again using set notation we would write $B = \{2, 4, 6\}$. The event $A \cup B$ is then the set of all outcomes such that the outcome is in A or in B or in both or these sets. In such simple examples it is sometimes helpful to depict the situation graphically.

The sample space $\mathcal{S}$ is the gray set containing all possible outcomes of our random experiment. Graphically the union of A and B, $A \cup B$ is a subset of the sample space, the entire colored area.

By the way: R provides functions for computing set operations. Let us use the occasion to show you briefly how to use these functions in the context of this example: We define the sets A and B first using the assignment operator:

```
A <- c(1,2,3)
B <- c(2,4,6)
```

We compute the union by using the function `union()`

```
union(A,B)
```

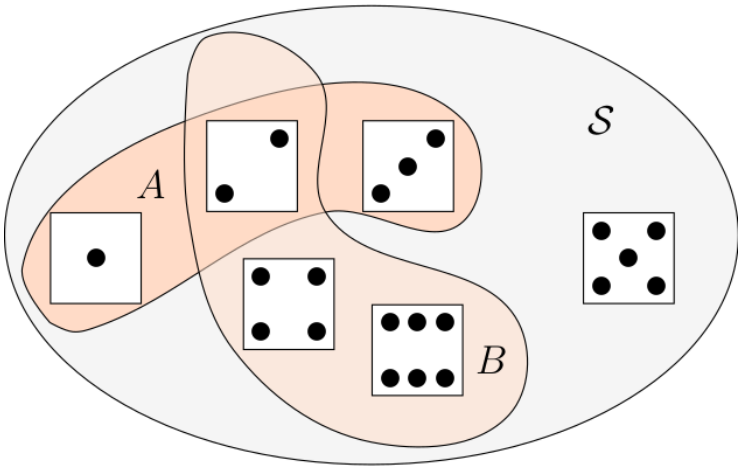

Figure 6: The meaning of set union

```
## [1] 1 2 3 4 6
```

which gives us the result we have already derived graphically.

Intersection The **intersection of two events** is the set that contains all events that are both in A and in B. Set intersection is written as $A \cap B$.

We can again illustrate this concept graphically

The intersection of A and B, $A \cap B$ is the orange area containing the dice face with two points. Indeed two is both in A and in B, which is exactly the meaning of set intersection.

The R function for computing intersections is calles `intersect()`. We call this function for our example

```
intersect(A,B)
```

```
## [1] 2
```

which gives us the result we have already derived graphically. Complement : The **complement** of an event A with respect to an event B is the set of all elements that are in B but not in A. Set difference is written as $A \setminus B$.

Lets say we want to get the complement, or the set difference of $A \cap B$ with respect to the sample space $\mathcal{S}$. Let us show $\mathcal{S} \setminus (A \cup B)$.

This complement is the dice shown in the light yellow area, i.e. all the elements of $\mathcal{S}$ which are not in $A \cup B$.

The R function for computing set complements is called `setdiff()`. You can

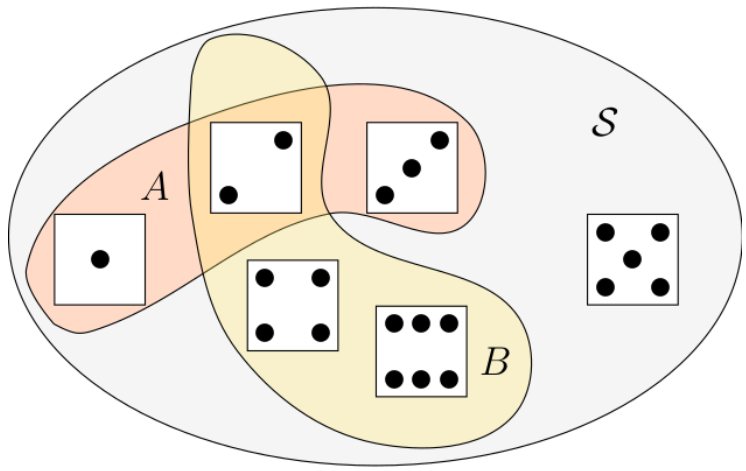

Figure 7: The meaning of set intersection

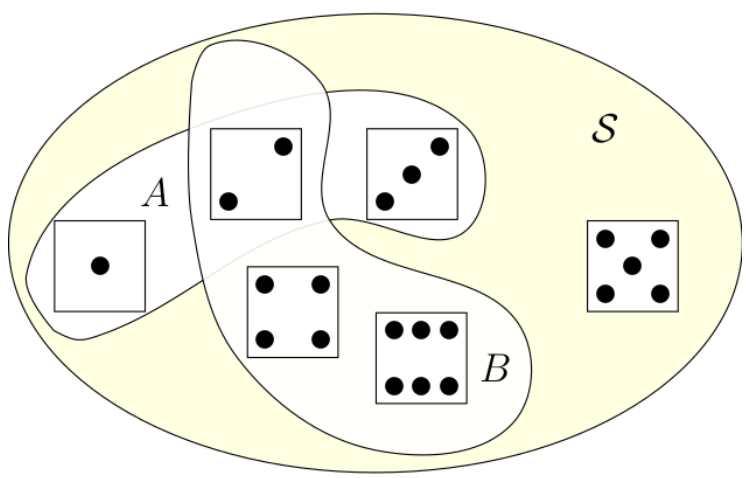

Figure 8: The meaning of set intersection

try it with our example:

```
S <- c(1,2,3,4,5,6)

setdiff(S, union(A,B))
```

```
## [1] 5
```

which is 5, as expected.

Mutually Exclusive Two events A and B are said to be **mutually exclusive** if they have no basic outcomes in common. The notation is $A \cap B = \emptyset$

An example in our context is the set of even outcomes $B = \{2, 4, 6\}$ and the set of odd outcomes, let us call it $C = \{1, 3, 5\}$. If we intersect these sets

```
B <- c(2,4,6)
C <- c(1,3,5)

intersect(B,C)
```

```
## numeric(0)
```

we get the empty set, which is expressed by R by giving the data type, in this case numeric, because we are intersecting sets of numeric values, followed by (0). This means, there is no numeric value in the intersection of B and C.

Let us discuss probability rule 3 a bit further. Let us go to the picture we drew to illustrate the meaning of set union @ref(fig:set-union). To compute the probability $P(A \cup B)$ we have to add the probabilities of all sample points that are contained either in A or in B but each point is to be counted only once. Therefore, probability rule 3, has a (weak) inequality: $P(A \cup B) \leq P(A) + P(B)$.

Now let E be a point contained both in A and in B, in our example this would be $E = \{2\}$, then $P(E)$ occurs twice on the right hand side of our inequality but only once on the left. Therefor the right hand side exceeds the left by the amount $P(A \cap B)$. Thus we have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

This reasoning holds for arbitrary pairs of events and not only in our example. In the case A and B are arbitrary events and A and B are mutually exclusive, i.e. $A \cap B = \emptyset$ then our inequality in rule 3 becomes an equality and we have $P(A \cup B) = P(A) + P(B)$.

Interpretations of Probability

While rule 1 - 3 pins down three basic mathematical properties of probabilities, there is an ongoing and still unsettled debate what probabilities actually mean. While in this course we will be less concerned with this philosophical question, it is so important that you hear about the basic interpretations at least once.

Probability theory took shape in the 17th century through discussions on the mathematics of games of chance, such as poker or roulette or gamed where dice are rolled. In this stage the *classical* meaning of probability was the ratio of outcomes that satisfy a condition or the description of an event divided by the total number of outcomes in the state space. Of course this interpretation of probability makes the implicit assumption that all outcomes have the same probability. For example, in this interpretation the probability that one role of a dice will result in a 6 is 1/6.

A more important interpretation is the interpretation of probability as *relative frequency*. In this interpretation the probability of an event A is interpreted as the number of times the event A occurs in repeated trials divided by the total number of trials in a random experiment.

A competing interpretation of probability is called *subjective probability*. In this interpretation probability is an opinion or a belief about how likely the occurence of a particular event is.

Examples

Methods

We describe our methods in this chapter.

Applications

Some *significant* applications are demonstrated in this chapter.

Example one

Example two

Final Words

We have finished a nice book.

Appendix

Installing R

Installing RStudio